# LESSON PLAN

Academic Year: 2026-27 | Programme: B.Tech

Semester: VII | Section: A,B,C,D

Name of the Course: Deep Learning | Course Code: V23CST15

**Course Outcomes (Along with Knowledge Level):**

| CO No. | Course Outcome | BTKL |
|---|---|---|
| 1 | Explain the fundamentals of Machine Learning and Deep Learning, including data preprocessing and evaluation techniques. | K2 |
| 2 | Illustrate the working and training process of Artificial Neural Networks and Deep Neural Networks with optimization techniques | K2 |
| 3 | Develop ANN models for classification tasks using Keras and model building pipeline | K3 |
| 4 | Develop CNNs and RNNs using PyTorch for solving real-world problems | K3 |
| 5 | Illustrate applications of Deep Learning in computer vision, natural language processing and sequence models | K2 |

## Text Books:

1. Deep Learning- Ian Goodfellow, YoshuaBengio and Aaron Courvile, MIT Press, 2016
2. Deep Learning with Python - Francois Chollet, Released December 2017, Publisher(s): Manning Publications, ISBN: 9781617294433
3. Deep Learning Illustrated: A Visual, Interactive Guide to Artificial Intelligence - Jon Krohn, Grant Beyleveld, AglaéBassens, Released September 2019, Publisher(s): Addison-Wesley Professional, ISBN: 9780135116821
4. Deep Learning from Scratch - Seth Weidman, Released September 2019, Publisher(s): O'Reilly Media, Inc., ISBN: 9781492041412

## Reference Books:

1. Artificial Neural Networks, Yegnanarayana, B., PHI Learning Pvt. Ltd, 2009.
2. Matrix Computations, Golub, G.,H., and Van Loan,C.,F, JHU Press,2013.
3. Neural Networks: A Classroom Approach, Satish Kumar, Tata McGraw-Hill Education, 2004.

**Targeted Proficiency Level (For each course Outcome):**

| COs | | CO1 | CO2 | CO3 | CO4 | CO5 |
|---|---|---|---|---|---|---|
| Targeted Proficiency Level | | 60 | 60 | 60 | 60 | 60 |
| Targeted level of Attainment | Level 3 | 60 | 60 | 60 | 60 | 60 |
| | Level 2 | 55 | 55 | 55 | 55 | 55 |
| | Level 1 | 50 | 50 | 50 | 50 | 50 |

**Lesson Plan:**

| S.No | Course Outcome | Intended Learning Outcomes (ILO) | K L of ILO | Hrs | Pedagogy | Teaching aids |
|---|---|---|---|---|---|---|
| 1 | CO 1 | Dissemination of Vision, Mission of the Dept. and PEOs, Pos, & PSOs of the Programme | | 1 | Lecture | ICT |
| 2 | | Demonstrate Artificial Intelligence, History of Machine learning | K2 | 1 | Lecture | ICT |
| 3 | | Explain the Probabilistic Modeling, Early Neural Networks, Kernel Methods | K2 | 2 | Lecture | ICT<br>Chalk & Talk |
| 4 | | Illustrate of Decision Trees, Random forests and Gradient Boosting Machines | K2 | 2 | Lecture with Discussion | ICT<br>Chalk & Talk |
| 5 | | Demonstrate the Fundamentals of Machine Learning: Four Branches of Machine Learning, Evaluating Machine learning Models, Overfitting and Underfitting | K2 | 3 | Lecture with Discussion | ICT |
| 6 | CO 2 | Explain Biological and Machine Vision | K2 | 2 | Lecture with Discussion | ICT<br>Chalk & Talk |
| 7 | | Illustrate Human and Machine Language | K2 | 1 | Lecture with Discussion | ICT<br>Chalk & Talk |
| 8 | | Explain Artificial Neural Networks | K2 | 3 | Lecture with Discussion | ICT |
| 9 | | Outline the Training Deep Networks, | K2 | 2 | Lecture with Discussion | ICT<br>Chalk & Talk |
| 10 | | Explain Improving Deep Networks | K2 | 2 | Lecture with Discussion | ICT<br>Chalk & Talk |
| 11 | CO 3 | Construct the Anatomy of Neural Network, Introduction to Keras; | K3 | 2 | Lecture with Discussion | ICT<br>Chalk & Talk |
| 12 | | Choose Keras, TensorFlow, | K3 | 2 | Lecture with Discussion | ICT<br>Chalk & Talk |

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| 13 | | Choose Theano and CNTK, | K3 | 2 | Lecture with Discussion | ICT Chalk & Talk |
| 14 | | Develop the Setting up Deep Learning Workstation, | K3 | 2 | Lecture with Discussion | ICT |
| 15 | | Develop Classifying Movie Reviews: Binary Classification | K3 | 3 | Lecture with Discussion | ICT |
| 16 | | Develop Classifying newswires: Multiclass Classification. | K3 | 3 | Lecture | ICT Chalk & Talk |
| 17 | | Organized talk about the unit | K3 | 2 | Lecture | ICT Chalk & Talk |
| 18 | CO 4 | Build a Neural Network and Representation Learning by incorporating ethics | K3 | 2 | Lecture | ICT Chalk & Talk |
| 19 | | Construct a Convolutional Layers, Multichannel Convolution Operation | K3 | 2 | Lecture | ICT Chalk & Talk |
| 20 | | Build Recurrent Neural Networks: Introduction to RNN | K3 | 2 | Lecture with Discussion | ICT Chalk & Talk |
| 21 | | Apply RNN Code & discuss it with respect to ethical point of view | K3 | 2 | Lecture with Discussion | ICT Chalk & Talk |
| 22 | | Apply PyTorch Tensors: Deep Learning with PyTorch | K3 | 2 | Lecture with Discussion | ICT Chalk & Talk |
| 23 | | Model of CNN in PyTorch with ethics | K3 | 1 | Lecture with Discussion | ICT Chalk &Talk |
| 24 | CO 5 | Demonstrate the Machine Vision, Natural Language processing | K2 | 2 | Lecture | ICT |
| 25 | | Explain Generative Adversial Networks along with its ethics | K2 | 2 | Lecture with Discussion | ICT Chalk &Talk |
| 26 | | Explain Deep Reinforcement Learning | K2 | 3 | Lecture with Discussion | ICT Chalk &Talk |
| 27 | | Demonstrate Deep Learning Research: Auto encoders | K2 | 3 | Lecture with Discussion | ICT Chalk &Talk |
| 28 | | Explain Deep Generative Models: Boltzmann Machines | K2 | 3 | Lecture with Discussion | ICT Chalk &Talk |
| 29 | | Demonstrate Deep Belief Networks added ethics | K2 | 1 | Lecture with Discussion | ICT |

**Total No. of Classes: 60**

**CO- PO & CO-PSO matrix:**

| COs/POs | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 1 | 1 | 1 | 1 | | | | | | 2 | 1 | |
| CO2 | 1 | 1 | 1 | 2 | 2 | | | | | | 2 | 1 | |
| CO3 | 1 | 1 | 2 | 2 | 3 | | | | | | 2 | 1 | 1 |
| CO4 | 1 | 2 | 2 | 2 | 2 | 1 | | | | | 2 | 1 | 1 |
| CO5 | 2 | 2 | 1 | 1 | 2 | 1 | | | | | 3 | 1 | 1 |
| Course | 1 | 1 | 1 | 1 | 2 | 1 | | | | | 2 | 1 | 1 |

**CO-PO & PSO Mapping Justification:**

| CO | CO-PO & PSO Mapping Justification | Justification |
|---|---|---|
| **CO1 (K2)** | **PO1 (2), PO2 (1), PO3 (1), PO4 (1), PO5 (1), PO11 (2), PSO1 (1), PSO2 (1)** | Acquire fundamental knowledge of **Machine Learning and Deep Learning**, explain AI concepts, data preprocessing techniques, train-validation-test split, and model evaluation metrics. Interpret ML model performance, recognize overfitting and underfitting issues, communicate foundational AI concepts effectively, and develop basic analytical skills required for intelligent systems. |
| **CO2 (K2)** | **PO1 (1), PO2 (1), PO3 (1), PO4 (2), PO5 (2), PO11 (2), PSO1 (1), PSO2 (1)** | Explain the architecture and working principles of **Artificial Neural Networks (ANNs)** and **Deep Neural Networks (DNNs)**, describe activation functions, backpropagation, optimization algorithms, weight initialization, and generalization techniques. Interpret network training processes using modern deep learning frameworks while understanding computational resource utilization. |
| **CO3 (K3)** | **PO1 (1), PO2 (1), PO3 (2), PO4 (2), PO5 (3), PO11 (2), PSO1 (1), PSO2 (2)** | Develop and implement **ANN-based classification models using Keras**, perform data preprocessing, model training, evaluation, and deployment basics. Analyze classification performance, design efficient neural network pipelines, employ modern deep learning tools, and build practical machine learning solutions for real-world datasets. |
| **CO4 (K3)** | **PO1 (1), PO2 (2), PO3 (2), PO4 (2), PO5 (2), PO6 (1), PO11 (2), PSO1 (1), PSO2 (2)** | Design and implement **Convolutional Neural Networks (CNNs)** and **Recurrent Neural Networks (RNNs)** using PyTorch. Analyze |

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| | | image and sequential data, apply convolution and recurrent learning concepts, evaluate deep learning models for real-world applications, and utilize modern frameworks while considering professional and ethical responsibilities in AI development. |
| **CO5 (K2)** | **PO1 (2), PO2 (2), PO3 (1), PO4 (1), PO5 (2), PO6 (1), PO11 (3), PSO1 (1), PSO2 (2)** | Explain the applications of **Deep Learning** in Computer Vision, Natural Language Processing, LSTM, GANs, Reinforcement Learning, Autoencoders, and real-world domains such as healthcare and agriculture. Interpret emerging AI technologies, recognize project management aspects of intelligent systems, and communicate the societal impact of deep learning applications. |

**Course End Survey Questionnaire:**

| S. No. | COs | Question |
|---|---|---|
| 1. | CO1 | How confident are you in distinguishing Machine Learning from Deep Learning and selecting appropriate evaluation metrics (Accuracy, Precision, Recall, and F1-score) for a given problem? |
| 2. | CO2 | How well did you understand the concepts of Artificial Neural Networks, back propagation, activation functions, and optimization techniques covered in this course? |
| 3. | CO3 | How confident are you in developing and training a neural network model using Keras by following the complete model-building pipeline (preprocessing, training, evaluation, and deployment basics)? |
| 4. | CO4 | To what extent are you able to differentiate between CNNs and RNNs and apply them using PyTorch for solving image and sequence-based problems? |
| 5. | CO5 | How effectively did the course help you understand the real-world applications of Deep Learning in Computer Vision, Natural Language Processing, LSTM, and other advanced topics? |

**Details of Course Instructors:**

| S.No. | Name of Course Instructor with designation | Sem/ Section | Contact No. & e-mail: | Signature of Course Instructor |
|---|---|---|---|---|
| 1 | Dr. A. Daveedu Raju<br>Professor | VII/A,B,C | 9866952163<br>davidrecharge@gmail.com | |
| 2 | Mr. E. Hanuman Sai Gupta<br>Assistant Professor | VII/D | 7989196435<br>saigupta.cse@srivasaviengg.ac.in | |

**Name of the Course Coordinator (with designation):** Mr. E. Hanuman Sai Gupta,
Assistant Professor

**Signature of the Course Coordinator:**

**Signature of the Module Coordinator:**

**Signature of the Head of the Department:**